

Supplementary Material for OntoCheck: Query-Driven Ontology Assessments for Scientific Domain Applications

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1 Complete Overview of Competency-Question Evaluation Tables

Tables SA–SF summarize the competency-question–based evaluation framework and ontology assessment results across multiple scientific domains. Table SA presents the domain-specific competency questions used to evaluate six ontologies spanning Capacitors, EBSD, GeoOutage, Geospatial, XRD, and Materials Processing applications (all of which can be subsetted from MDS-Onto). Table SB reports the per-question task-based ontology performance metrics, including ontology term counts, matched task terms, relevance, and accuracy for each competency question. Table SC summarizes the aggregate task-based assessment results across all evaluated domains. Table SD defines the generalized competency-question benchmark used for cross-domain materials science ontology evaluation, organized into the following categories: Material Identity, Manufacturing, Characterization, Properties, Provenance, Simulation, and Degradation. Table SE reports detailed competency-question coverage scores for each evaluated ontology (MDS-Onto, PMDCo, AM-Ontology, EMMO, and CHAMEO), including category-wise subtotals and overall averages. Finally, Table SF summarizes the overall ontology coverage performance and ontology scale in terms of total ontology terms and aggregate competency-question coverage percentages across all evaluated materials science ontologies.

Table SA: Domain competency questions (CQ) for each ontology.

CQ	Question
Capacitors Ontology	
1	What types of electrical connectors are available in the EC-HALT system, and what is the function of each?

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CQ Question	
2	What is the wire gauge and material of the electrical wire used in the HALT testing setup?
3	What equipment components are classified as part of the EC-HALT?
4	What is the phase-to-ground voltage rating of the electrical wire used in the testing system?
5	What type of insulation material is used on the electrical wires, and what is its diameter?
6	What sensor is used to monitor relative humidity inside the exposure chamber?
7	What is the height setting of the wire grate within the exposure chamber, and how many wire grates are in use?
8	What type of thermocouple is used in the exposure chamber, and what is its operational temperature range?
9	What are the manufacturer-specified dimensions (height, length, and width) of the capacitor under test?
10	What is the capacitance value and its tolerance for the capacitor being tested?
11	What are the upper and lower temperature limits specified by the manufacturer for the capacitor?
12	What is the voltage rating of the capacitor, and what application norm is it rated for?
13	What is the lot code of the capacitor, and who is the supplier?
14	What type of capacitor is being tested—is it a MultiLayer Ceramic Capacitor (MLCC) or another type?
15	What is the packaging material the capacitor was supplied in?
16	What are the terminal electrode length and its tolerance for the capacitor under test?
17	What is the design and coating type of the printed circuit board (PCB) used as the sample holder?
18	How many layers does the PCB have, and what is its glass transition temperature?
19	What solder process type and solder material were used when mounting the capacitor onto the PCB, and at what iron temperature?
20	Was external flux used during the soldering process, and if so, what type of flux material was applied?
EBSO Ontology	
1	What is the material composition of this sample?
2	What laser power was used for this sample?
3	What manufacturing method was used to produce this sample?
4	What surface of this sample was prepared for EBSD characterization?
5	What is the total grinding removal depth of this sample?
6	What is the mass change recorded in the surface quality check of this sample?
7	Who performed the grinding operation and on what date was it carried out?

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CQ Question

- 8 What solvent and drying gas were used in the cleaning step of this sample's processing?
 - 9 What equipment was used in the grinding step of this sample's preparation?
 - 10 What is the polishing voltage used in the dry electropolishing of this sample?
 - 11 What is the grinding wheel speed of the machine used in the grinding step of this sample's processing?
 - 12 What coolant is delivered by the grinding machine used in the grinding step of this sample's processing?
 - 13 What is the relative humidity of the electrolyte bead media used in the electropolish system for this sample's dry electropolishing?
 - 14 What cathode material is a component of the electropolish system used in the dry electropolishing step of this sample's preparation?
 - 15 For this sample, what are both the grinding wheel speed and the depth of cut per pass of the grinding machine used in its preparation?
 - 16 For this sample, what are the polishing voltage, motor speed setting, and frequency mode of the electropolish system used in its dry electropolishing?
 - 17 For this sample, what are the grinding wheel speed used during grinding and the polishing voltage used during dry electropolishing?
 - 18 Which samples have a grinding-induced deformation layer depth greater than 50 micrometers?
 - 19 Which samples have the highest electropolish rate of mass change?
 - 20 Which samples from the same batch that were manufactured by LPBF with a laser power above 200W had a total surface removal depth less than 100 micrometers after preparation?
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GeoOutage Ontology

- 1 Which counties have the highest number of outages?
 - 2 What are the top 100 outage records by number of outages?
 - 3 Which outage records were recorded in 2022?
 - 4 Which NTL images represent Orange County, Florida in 2025?
 - 5 Which county does this outage record represent?
 - 6 Which county does this NTL image represent?
 - 7 Which county does this outage map represent?
 - 8 What is the record date and time of this outage record?
 - 9 How many outages were recorded in this outage record?
 - 10 Which NTL image is associated with this outage record?
 - 11 Which outage map is associated with this outage record?
 - 12 Which outage map was generated from this NTL image?
 - 13 Which satellite captured this NTL image?
 - 14 Which sensor captured this NTL image?
 - 15 Which outage records in Lee County, Florida had more than 100 outages around Hurricane Ian?
 - 16 Which NTL images are linked to outage records in Lee County, Florida around Hurricane Ian?
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CQ Question

- 17 Which outage maps are linked to outage records in Lee County, Florida around Hurricane Ian?
 - 18 For each county, what is the maximum number of outages recorded?
 - 19 Which NTL image and outage map correspond to the highest-outage record for each county?
 - 20 On what date and time did each county reach its maximum recorded number of outages?
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Geospatial Ontology

- 1 How many geohashes exist within a certain bounding box in EPSG:4326 coordinate reference system?
 - 2 What satellite is the ASTER sensor mounted on?
 - 3 How many bands does the ASTER sensor collect?
 - 4 What of the available geospatial data has a spatial resolution of 250 meters or lower (i.e., finer)?
 - 5 What geospatial raster data contains information about elevation?
 - 6 What geospatial raster data was acquired in 2016?
 - 7 What geospatial raster data exists in the geohash “dpn”?
 - 8 What geohashes border the geohash “bccc”?
 - 9 What bands contain near infrared information from the ASTER sensor?
 - 10 What satellites were launched in 2020?
 - 11 What geospatial data exists in the EPSG:5070 coordinate reference system?
 - 12 What geospatial data exists in the neighborhood of geohash “dpnq”?
 - 13 What satellites are still operational?
 - 14 What satellites and sensors are owned by NASA?
 - 15 What geotiff files are available from 2020 to 2022?
 - 16 What raster datasets are available from the Terra satellite?
 - 17 What band categories are available for the ASTER sensor?
 - 18 What raster data is available every 30 minutes in the geohash “bcff”?
 - 19 What is the finest spatial resolution available from Terra datasets?
 - 20 What is the finest temporal resolution available from Terra datasets?
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XRD Ontology

- 1 What synchrotron facility is this beamline part of?
 - 2 What X-ray detector was used in this experiment?
 - 3 What is the pixel size width of this detector?
 - 4 What processing method was used to manufacture this sample?
 - 5 What is the chemical formula of this sample?
 - 6 What are the lattice parameters determined by Rietveld refinement of this diffraction pattern?
 - 7 What is the dislocation density obtained from diffraction line profile analysis?
 - 8 What phase fractions were determined by Rietveld refinement?
 - 9 What is the beta transus temperature for this titanium alloy sample?
 - 10 What crystal structure does the alpha-titanium phase have?
 - 11 What grain morphology results from rapid solidification in LPBF?
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CQ	Question
12	What thermal profile was prescribed for this heat treatment?
13	What is the cooling rate used in this continuous cooling transformation experiment?
14	What azimuthal integration parameters were used to produce this 1D diffraction pattern?
15	What types of solid-state phase transformations are defined in the ontology?
16	What beamlines are available at the Advanced Photon Source?
17	What governing laws relate lattice parameters to other measurable quantities?
18	What phase does the beta-titanium phase transform into on cooling?
19	What is the isothermal hold temperature prescribed for this transformation experiment?
20	What additive manufacturing processes are defined in the ontology and what solidification behavior do they promote?
Materials Processing (MatProc) Ontology	
1	What are all materials processing techniques in the ontology?
2	Which processes are subtypes of Sintering?
3	Which processes are subtypes of Welding?
4	Which processes are subtypes of Casting?
5	What parameters are required for Laser Powder Bed Fusion?
6	What parameters are required for Spark Plasma Sintering?
7	Which processes require ShieldingGas as a parameter?
8	Which processes require Wavelength as a parameter?
9	Which processes declare no processing parameters?
10	How many parameters does each process declare? (ranked)
11	What are all subtypes of ProcessingTemperature?
12	What are all subtypes of DwellTime?
13	What are all subtypes of ProcessingPressure?
14	What unit is associated with ScanSpeed?
15	Which parameters are measured in degrees Celsius?
16	Which parameters have no unit declared?
17	Which processes cite an ISO or ASTM standard?
18	Which process is known by the abbreviation "SPS"?
19	Which process is known by the abbreviation "HEBM"?
20	Which processes share both a temperature parameter and a pressure parameter?

Table SB: Per-question task-based ontology performance. $|T_o|$: ontology terms; $|T_a|$: task terms extracted from the SPARQL query; $|T_a \cap T_o|$: matched terms; Relevance (Recall) = $|T_a \cap T_o|/|T_a|$; Accuracy (Precision) = $|T_a \cap T_o|/|T_o|$.

Domain	CQ	$ T_o $	$ T_a $	$ T_a \cap T_o $	Relevance	Accuracy
EBSD Ontology						
EBSD	1	89	2	1	50%	1.1%
EBSD	2	89	2	1	50%	1.1%
EBSD	3	89	2	1	50%	1.1%
EBSD	4	89	2	1	50%	1.1%
EBSD	5	89	2	1	50%	1.1%
EBSD	6	89	3	3	100%	3.4%
EBSD	7	89	3	3	100%	3.4%
EBSD	8	89	5	5	100%	5.6%
EBSD	9	89	5	5	100%	5.6%
EBSD	10	89	6	6	100%	6.7%
EBSD	11	89	6	6	100%	6.7%
EBSD	12	89	6	6	100%	6.7%
EBSD	13	89	8	8	100%	9.0%
EBSD	14	89	7	7	100%	7.9%
EBSD	15	89	7	7	100%	7.9%
EBSD	16	89	8	8	100%	9.0%
EBSD	17	89	9	9	100%	10.1%
EBSD	18	89	2	1	50%	1.1%
EBSD	19	89	2	1	50%	1.1%
EBSD	20	89	7	6	86%	6.7%
Avg.		89	—	—	82%	4.8%
Capacitors Ontology						
Capacitors	1	87	1	1	100%	1.1%
Capacitors	2	87	3	3	100%	3.4%
Capacitors	3	87	1	1	100%	1.1%
Capacitors	4	87	1	1	100%	1.1%
Capacitors	5	87	2	2	100%	2.3%
Capacitors	6	87	1	1	100%	1.1%
Capacitors	7	87	2	2	100%	2.3%
Capacitors	8	87	1	1	100%	1.1%
Capacitors	9	87	4	4	100%	4.6%
Capacitors	10	87	3	3	100%	3.4%
Capacitors	11	87	3	3	100%	3.4%
Capacitors	12	87	3	3	100%	3.4%
Capacitors	13	87	3	3	100%	3.4%
Capacitors	14	87	1	1	100%	1.1%
Capacitors	15	87	2	2	100%	2.3%
Capacitors	16	87	3	3	100%	3.4%
Capacitors	17	87	3	3	100%	3.4%
Capacitors	18	87	3	3	100%	3.4%

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Domain	CQ	$ T_o $	$ T_a $	$ T_a \cap T_o $	Relevance	Accuracy
Capacitors	19	87	4	4	100%	4.6%
Capacitors	20	87	3	3	100%	3.4%
Avg.		87	—	—	100%	2.7%
GeoOutage Ontology						
GeoOutage	1	10	3	3	100%	30.0%
GeoOutage	2	10	4	4	100%	40.0%
GeoOutage	3	10	3	3	100%	30.0%
GeoOutage	4	10	2	2	100%	20.0%
GeoOutage	5	10	2	2	100%	20.0%
GeoOutage	6	10	2	2	100%	20.0%
GeoOutage	7	10	2	2	100%	20.0%
GeoOutage	8	10	2	2	100%	20.0%
GeoOutage	9	10	2	2	100%	20.0%
GeoOutage	10	10	2	2	100%	20.0%
GeoOutage	11	10	2	2	100%	20.0%
GeoOutage	12	10	2	2	100%	20.0%
GeoOutage	13	10	2	2	100%	20.0%
GeoOutage	14	10	2	2	100%	20.0%
GeoOutage	15	10	4	4	100%	40.0%
GeoOutage	16	10	6	6	100%	60.0%
GeoOutage	17	10	6	6	100%	60.0%
GeoOutage	18	10	3	3	100%	30.0%
GeoOutage	19	10	5	5	100%	50.0%
GeoOutage	20	10	4	4	100%	40.0%
Avg.		10	—	—	100%	30.0%
Geospatial Ontology						
Geospatial	1	100	7	7	100%	7.0%
Geospatial	2	100	4	4	100%	4.0%
Geospatial	3	100	3	3	100%	3.0%
Geospatial	4	100	2	2	100%	2.0%
Geospatial	5	100	3	3	100%	3.0%
Geospatial	6	100	2	2	100%	2.0%
Geospatial	7	100	3	3	100%	3.0%
Geospatial	8	100	9	9	100%	9.0%
Geospatial	9	100	5	5	100%	5.0%
Geospatial	10	100	3	3	100%	3.0%
Geospatial	11	100	3	3	100%	3.0%
Geospatial	12	100	11	11	100%	11.0%
Geospatial	13	100	3	3	100%	3.0%
Geospatial	14	100	6	6	100%	6.0%
Geospatial	15	100	3	3	100%	3.0%
Geospatial	16	100	4	4	100%	4.0%
Geospatial	17	100	4	4	100%	4.0%
Geospatial	18	100	4	4	100%	4.0%
Geospatial	19	100	4	4	100%	4.0%

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Domain	CQ	$ T_o $	$ T_a $	$ T_a \cap T_o $	Relevance	Accuracy
Geospatial	20	100	4	4	100%	4.0%
	Avg.	100	—	—	100%	4.4%
MatProc Ontology						
MatProc	1	127	1	1	100%	0.8%
MatProc	2	127	1	1	100%	0.8%
MatProc	3	127	1	1	100%	0.8%
MatProc	4	127	1	1	100%	0.8%
MatProc	5	127	2	2	100%	1.6%
MatProc	6	127	2	2	100%	1.6%
MatProc	7	127	2	2	100%	1.6%
MatProc	8	127	2	2	100%	1.6%
MatProc	9	127	2	2	100%	1.6%
MatProc	10	127	1	1	100%	0.8%
MatProc	11	127	1	1	100%	0.8%
MatProc	12	127	1	1	100%	0.8%
MatProc	13	127	1	1	100%	0.8%
MatProc	14	127	1	1	100%	0.8%
MatProc	15	127	1	1	100%	0.8%
MatProc	16	127	1	1	100%	0.8%
MatProc	17	127	1	1	100%	0.8%
MatProc	18	127	0	0	0%	0.0%
MatProc	19	127	0	0	0%	0.0%
MatProc	20	127	3	3	100%	2.4%
	Avg.	127	—	—	90%	1.0%
XRD Ontology						
XRD	1	318	2	2	100%	0.6%
XRD	2	318	2	2	100%	0.6%
XRD	3	318	2	2	100%	0.6%
XRD	4	318	2	2	100%	0.6%
XRD	5	318	2	2	100%	0.6%
XRD	6	318	2	2	100%	0.6%
XRD	7	318	2	2	100%	0.6%
XRD	8	318	2	2	100%	0.6%
XRD	9	318	2	2	100%	0.6%
XRD	10	318	3	3	100%	0.9%
XRD	11	318	2	2	100%	0.6%
XRD	12	318	2	2	100%	0.6%
XRD	13	318	2	2	100%	0.6%
XRD	14	318	2	2	100%	0.6%
XRD	15	318	1	1	100%	0.3%
XRD	16	318	1	1	100%	0.3%
XRD	17	318	2	2	100%	0.6%
XRD	18	318	2	2	100%	0.6%
XRD	19	318	2	2	100%	0.6%
XRD	20	318	3	3	100%	0.9%

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Domain	CQ	$ T_o $	$ T_a $	$ T_a \cap T_o $	Relevance	Accuracy
	Avg.	318	—	—	100%	0.6%

Table SC. Task-Based Ontology Assessment Results Across All Domains

Domain	$ T_o $	$ T_a $	$ T_a \cap T_o $	Relevance	Accuracy
EBSD	89	36	35	97.22%	39.33%
Capacitors	87	38	38	100.00%	43.68%
GeoOutage	10	10	10	100.00%	100.00%
Geospatial	100	37	37	100.00%	37.00%
Materials Processing	127	14	14	100.00%	11.02%
XRD	318	33	33	100.00%	10.38%

Table SD: Competency Questions for Materials Science Ontology Assessment

ID	Category	Competency Question
<i>Material Identity</i>		
Q01	Material Identity	What material samples are available in the dataset, and what is the material class (metal/ceramic/polymer, composite) of each?
Q02	Material Identity	What is the identifier/name assigned to each material specimen, and by whom was it assigned?
Q03	Material Identity	What is the physical state (s/l/g/powder/...) of each material sample at the time of processing or characterization?
Q04	Material Identity	What is the chemical composition of a given material specimen, including all constituent elements or compounds and their respective fractions?
Q05	Material Identity	What dimensional specifications (length, width, height, thickness) are recorded for a material specimen?
Q06	Material Identity	What is the mass or volume of a material specimen as recorded in the dataset?
Q07	Material Identity	What material samples share the same chemical composition but were produced by different manufacturing processes?

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Table SD continued

ID	Category	Competency Question
Q08	Material Identity	What is the aggregate state or phase (crystalline, amorphous, multiphase) of a material at a given stage of its lifecycle?
<i>Manufacturing</i>		
Q09	Manufacturing	What manufacturing or synthesis process was used to produce each material specimen?
Q10	Manufacturing	What process parameters were recorded during the manufacturing of a specimen, and what are their measured values and units?
Q11	Manufacturing	What temperature was applied during a thermal processing step (sintering, annealing, quenching, curing), and for how long?
Q12	Manufacturing	What raw input materials or feedstocks were consumed during the manufacturing process, and in what quantities?
Q13	Manufacturing	What intermediate processing steps occurred between raw material acquisition and the final specimen, and in what sequence?
Q14	Manufacturing	What tools or equipment were used during the manufacturing process, and what is the functional role of each?
Q15	Manufacturing	What were the environmental conditions (inert atmosphere, vacuum, humidity level) maintained during manufacturing?
Q16	Manufacturing	What post-processing or surface treatment was applied to a specimen after primary manufacturing, and what parameters were recorded?
<i>Characterization</i>		
Q17	Characterization	What characterization techniques were applied to a given specimen, and in what order were they performed?
Q18	Characterization	What instrument or device was used for each characterization measurement, and what is its model or identifier?
Q19	Characterization	What measurement parameters (voltage, current, scan rate, exposure time, wavelength) were set for a characterization experiment?
Q20	Characterization	What output or result was produced by each characterization measurement, and in what unit is it expressed?

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Table SD continued

ID	Category	Competency Question
Q21	Characterization	What specimen preparation steps were carried out before a characterization measurement was performed?
Q22	Characterization	What calibration procedure was applied to the measurement instrument before data acquisition, and what reference standard was used?
Q23	Characterization	What is the measurement uncertainty or tolerance associated with a reported characterization result?
Q24	Characterization	What environmental conditions (temperature, humidity, pressure) were maintained during the characterization measurement?
Q25	Characterization	What data format or file type was used to store the raw output of a characterization experiment?
Q26	Characterization	What analysis or data processing method was applied to raw characterization data to derive the final reported material property?
<i>Properties</i>		
Q27	Properties	What mechanical properties (tensile strength, yield strength, elastic modulus, hardness, fracture toughness) are reported for a given material?
Q28	Properties	What thermal properties (melting point, glass transition temperature, thermal conductivity, heat capacity) are recorded for a material specimen?
Q29	Properties	What electrical or dielectric properties (electrical conductivity, dielectric constant, permittivity) are reported for a material?
Q30	Properties	What optical properties (absorbance, transmittance, reflectivity, refractive index) are recorded for a material specimen?
Q31	Properties	What structural or microstructural properties (grain size, porosity, crystallographic texture, phase fraction) are reported for a material?
Q32	Properties	What property values exceed or fall below a specified threshold/limit defined by a standard (ASTM, ISO, etc.) or specification?
Q33	Properties	What is the reported density of a material specimen, and by what method was it determined?
Q34	Properties	What is the surface roughness or surface area of a material specimen, and at what scale was it measured?

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Table SD continued

ID	Category	Competency Question
<i>Provenance & Metadata</i>		
Q35	Provenance	Who performed the manufacturing or characterization process—which person, organization, or automated system is identified as the responsible agent?
Q36	Provenance	What institution or facility owns or operates the equipment used in a characterization or manufacturing experiment?
Q37	Provenance	What standard, specification, or norm does a given process or measurement comply with?
Q38	Provenance	On what date or within what time period was a manufacturing or characterization process carried out?
Q39	Provenance	What is the version or lot identifier of the material or reagent used in an experiment, and who supplied it?
Q40	Provenance	What is the purpose or objective of a given experiment or process—what material property or behavior was it intended to investigate?
<i>Simulation</i>		
Q41	Simulation	What simulation or computational modeling method was applied to a material or process, and what software or computing environment was used?
Q42	Simulation	What input parameters were supplied to a simulation, and what output quantities did it produce?
Q43	Simulation	What physical assumptions or simplifications were made in a computational model of a material or process?
Q44	Simulation	At what spatial or temporal scale does a given simulation operate, and is it atomistic, mesoscale, or continuum?
Q45	Simulation	How does a simulated material property compare to the experimentally measured value reported in the same dataset?
<i>Degradation & Lifetime</i>		
Q46	Degradation	What degradation or damage mechanisms (corrosion, fatigue, creep, oxidation, delamination) are represented for a material under service or test conditions?

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Table SD continued

ID	Category	Competency Question
Q47	Degradation	What stressor types (mechanical load, thermal cycling, UV radiation, humidity, chemical exposure) were applied to a material during a degradation or lifetime study?
Q48	Degradation	What measurable change in a material property (drop in tensile strength, increase in porosity, color shift) was recorded as a function of exposure time or number of cycles?
Q49	Degradation	What is the estimated or measured service lifetime of a material or component under specified operating conditions?
Q50	Degradation	What (primary) failure mode or end-of-life criterion was defined for a material/component, and was that criterion reached during the study period?

Table SE: Competency-Question Coverage Scores (%) for Each Ontology

ID	Category	MDS-Onto	PMDCo	AM-Onto	EMMO	CHAMEO
<i>Material Identity</i>						
Q01	Material Identity	100	100	67	100	33
Q02	Material Identity	100	100	0	100	33
Q03	Material Identity	100	100	100	100	100
Q04	Material Identity	100	100	100	100	67
Q05	Material Identity	100	100	100	100	50
Q06	Material Identity	100	100	100	100	100
Q07	Material Identity	100	100	100	100	100
Q08	Material Identity	100	100	100	100	67
Subtotal		100	100	83	100	69
<i>Manufacturing</i>						
Q09	Manufacturing	100	100	100	100	100
Q10	Manufacturing	100	100	100	100	75
Q11	Manufacturing	100	100	100	100	67
Q12	Manufacturing	100	100	100	100	67
Q13	Manufacturing	100	67	67	100	100
Q14	Manufacturing	100	100	100	100	67
Q15	Manufacturing	100	100	100	100	100
Q16	Manufacturing	100	100	100	100	100

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Table SE continued

ID	Category	MDS-Onto	PMDCo	AM-Onto	EMMO	CHAMEO
Subtotal		100	96	96	100	84
<i>Characterization</i>						
Q17	Characterization	100	100	100	100	100
Q18	Characterization	100	100	67	100	100
Q19	Characterization	100	100	67	100	100
Q20	Characterization	100	100	67	100	67
Q21	Characterization	100	100	0	100	100
Q22	Characterization	100	67	67	67	100
Q23	Characterization	100	100	0	100	50
Q24	Characterization	100	100	67	67	100
Q25	Characterization	100	100	67	100	100
Q26	Characterization	100	100	67	100	100
Subtotal		100	97	57	93	92
<i>Properties</i>						
Q27	Properties	100	100	100	100	100
Q28	Properties	100	100	100	100	67
Q29	Properties	100	100	67	100	100
Q30	Properties	100	100	100	67	100
Q31	Properties	100	100	100	100	33
Q32	Properties	100	100	100	100	100
Q33	Properties	100	100	100	100	67
Q34	Properties	100	100	100	100	67
Subtotal		100	100	96	96	79
<i>Provenance & Metadata</i>						
Q35	Provenance	67	100	33	100	67
Q36	Provenance	100	67	33	67	100
Q37	Provenance	67	67	67	100	67
Q38	Provenance	100	100	100	100	100
Q39	Provenance	100	100	67	100	67
Q40	Provenance	100	100	67	100	67
Subtotal		89	89	61	94	78
<i>Simulation</i>						
Q41	Simulation	100	100	67	67	67
Q42	Simulation	100	100	100	100	100
Q43	Simulation	67	67	100	67	67
Q44	Simulation	67	100	100	100	33
Q45	Simulation	100	100	33	100	67
Subtotal		87	93	80	87	67

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Table SE continued

ID	Category	MDS-Onto	PMDCo	AM-Onto	EMMO	CHAMEO
<i>Degradation & Lifetime</i>						
Q46	Degradation	67	100	33	100	67
Q47	Degradation	100	100	67	100	33
Q48	Degradation	100	100	100	100	67
Q49	Degradation	100	67	67	100	67
Q50	Degradation	100	67	67	100	33
Subtotal		93	87	67	100	53
Overall Average (%)		96.7	95.3	77.3	96.0	76.8

Table SF. Competency-Question Coverage of Materials Science Ontologies

	MDS-Onto	PMDCo	AM-Ontology	EMMO	CHAMEO
Terms	2,187	1,469	695	3,859	255
Coverage (%)	96.7	95.3	77.3	96.0	76.8

2 Further Resource Availability

All ontologies evaluated in this work are available through one of the following channels. Ontologies developed within the MDS-Onto ecosystem, including the EBSD, Capacitors, Materials Processing, XRD, GeoOutage, and Geospatial ontologies, are provided in the supplementary zip file accompanying this submission. Where applicable, these ontologies are organized into modular components reflecting their development structure, and can be subsetted directly from MDS-Onto. All other ontologies evaluated in the comparative assessment, MDS-Onto (full), PMDCo, AM-Onto, EMMO, and CHAMEO, have been cited in the main paper and are publicly accessible via their respective repositories.

All SPARQL queries corresponding to the competency questions reported in this work are included in the supplementary zip file. Additional resources, including the OntoCheck source code, installation instructions, and links to the published Python package, are available through the project GitHub repository at <https://github.com/cwru-sdle/OntoCheck>. OntoCheck is fully open-source and released under the CC BY 4.0 license.